

Significant stages in the evolution of mobile radio systems are referred to in terms of generations: 0th generation or 0G, first generation or 1G, second generation or 2G, third generation or 3G, and fourth generation or 4G. Reviewing previous wireless generations will help put 4G in context.

0G systems preceded modern mobile telephony and are referred to as mobile radio telephone systems. Mobile radio service originated with Bell Systems in 1946. Technologies used in pre-mobile systems included push-to-talk (PTT or manual), mobile telephone system (MTS), mobile telephony system A/B/C/D (MTA/B/C/D), *offentlig landmobil telefoni* (OLT), improved mobile telephone service (IMTS) and advanced mobile telephone system (AMTS).

The 1G of mobile generation came in 1980s, offering for the first time a relatively affordable mobile wireless telephony service. 1G was characterised by a multiplicity of incompatible regional analogue standards such as analogue advanced mobile phone system (AMPS)—North American standard in mobile band (800MHz), total access communications system (TACS)—the UK-originated standard based on AMPS in 900MHz band, Nordic mobile telephony system (NMT)—Scandinavian standard in 450MHz and 900MHz bands, C-450—German standard in 450MHz band and Japanese total access communications system (JTACS)—Japanese standard in 900MHz band. These incompatible analogue standards kept the market fragmented, expensive and without national roaming facility.

In early 1990s, global system for mobile (GSM) communication, the first of the digital 2G, arrived to provide telephony plus text messaging and limited circuit-switched data services. For GSM, everything lined up—technology, demand, supply, pricing, value and delivery costs. The result remains an enormous international industry that in 15 years

IMT-Advanced Requirements			
Criterion	Test environment		Threshold
Cell spectral efficiency (bit/s/Hz/cell) [The aggregate throughput of all users (the number of correctly received bits over a certain period of time) divided by channel bandwidth divided by the number of cells.]	Indoor	Downlink	3.0
		Uplink	2.25
	Micro-cellular	Downlink	2.6
		Uplink	1.8
	Urban coverage	Downlink	2.2
Peak spectral efficiency (bit/s/Hz) (Highest theoretical data rate)	High speed	Downlink	1.1
		Uplink	0.7
		Downlink	15.0
		Uplink	6.75
			40
Bandwidth (MHz) (Candidate technologies are supposed to support a scalable bandwidth upto 40MHz and are encouraged to operate upto 100MHz.)	Indoor	Downlink	0.1
		Uplink	0.07
	Micro-cellular	Downlink	0.075
		Uplink	0.05
	Urban coverage	Downlink	0.06
Cell edge user spectral efficiency (bit/s/Hz) [Five per cent point of cumulative distribution function (CDF) of the normalised user throughput.]	High speed	Downlink	0.04
		Uplink	0.015
	Control plane		<100
		User plane	<10
Latency (ms) (Transition from different connection modes, for example, from idle to active state.)	Indoor	@10km/h	1.0
		@30km/h	0.75
	Micro-cellular	@120km/h	0.55
		@350km/h	0.25
	High speed		0.25
Mobility (bit/s/Hz)	Indra-frequency HO interruption (ms)		27.5
		Inter-frequency HO interruption within a band (ms)	40
	Inter-frequency HO interruption between bands (ms)		60
			60
			60
Handover (HO) (The handover interruption time is defined as the time duration during which a user terminal cannot exchange user plane packets with any base station.)	Indoor		50
		Micro-cellular	40
	Micro-cellular	Urban coverage	40
		High speed	30
	High speed		30
Voice-over IP capacity (active/users/sector/MHz)	Indoor		50
		Micro-cellular	40
	Micro-cellular	Urban coverage	40
		High speed	30
	High speed		30

grew from nothing to being owned by more than half the people on the planet, revolutionising the way the world communicates.

2.5G, which stands for second and a half generation, is a mobile wireless technology developed in between its predecessor, 2G, and its successor, 3G. 2.5G is used to describe 2G systems that have implemented a packet-switched domain in addition to circuit-switched domain. Such a packet-switched base service

is general packet radio service (GPRS) that can provide moderate-speed data rates from 56kbps to 114kbps using time division multiple access (TDMA) channels in the GSM system.

Another extension of GSM is 2.75G—enhanced data rates for GSM evolution (EDGE) or IMT-SC (single carrier) that is able to support both circuit-switched and packet-switched networks. EDGE can carry a bandwidth up to 236.8kbps for four timeslots (theoretical maximum